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a photo of a giraffe and a banana



Stable Diffusion



Ours

a photo of a giraffe and a bear



Stable Diffusion



Ours

a photo of a car and a blue cat



Stable Diffusion

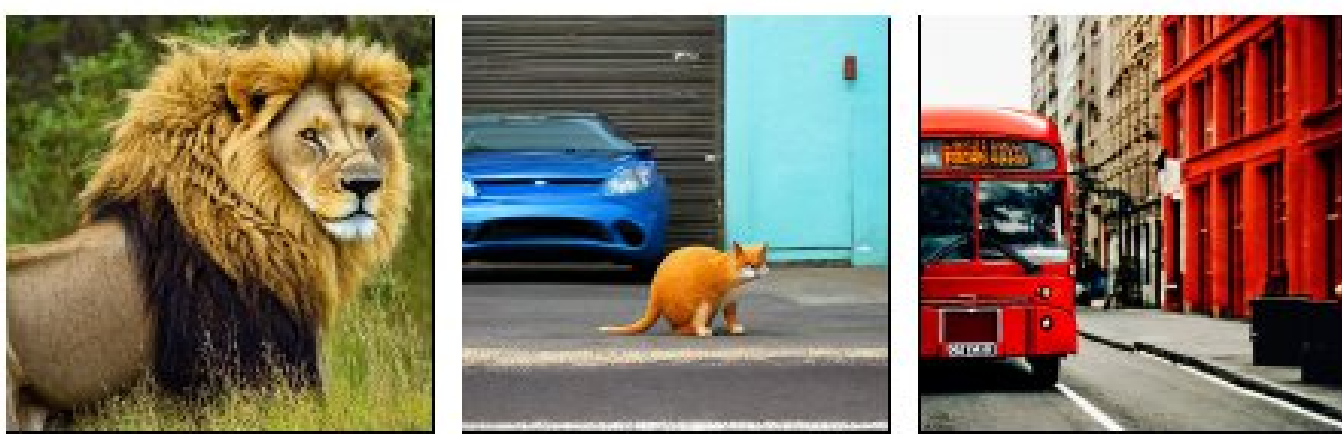


Ours

## Context and objective

**Context:** diffusion models generate good quality images from a prompt

**Limits:**



Catastrophic neglect \* "a photo of a lion and a bear"  
Attribute binding \* "a photo of a blue cat and a yellow car"  
Attribute linking \* "a photo of a red bus driving down the street"

**Objective:** improve text-image alignment

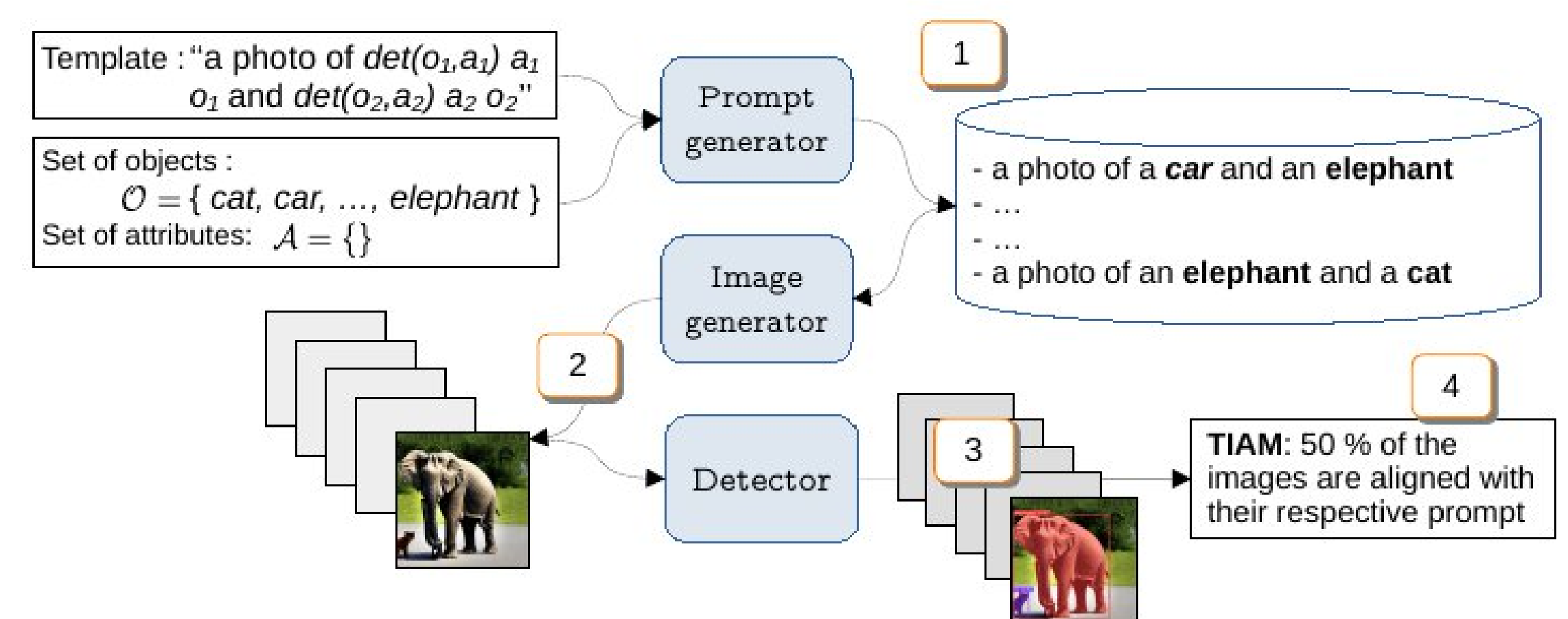


Fig 1. Measuring the alignment with TIAM [Grimal et al., WACV 2024]

## Devil is in the details

- Diffusion Model: learn to denoise images
  - gaussian noise  $\rightarrow$  denoise  $\rightarrow$  new image
- Structure (low frequencies) are built first
  - choosing the « denoising speed » is difficult
  - semantic guidance – *Generative Semantic Nursing* (GSN)
- InitNO [Guo et al., 2024]: use several initial noises
- Proposition: choose the best diffusion step for semantic guidance
  - determined with cross-validation on few (16) prompts
  - reduce the number of hyperparameters to set

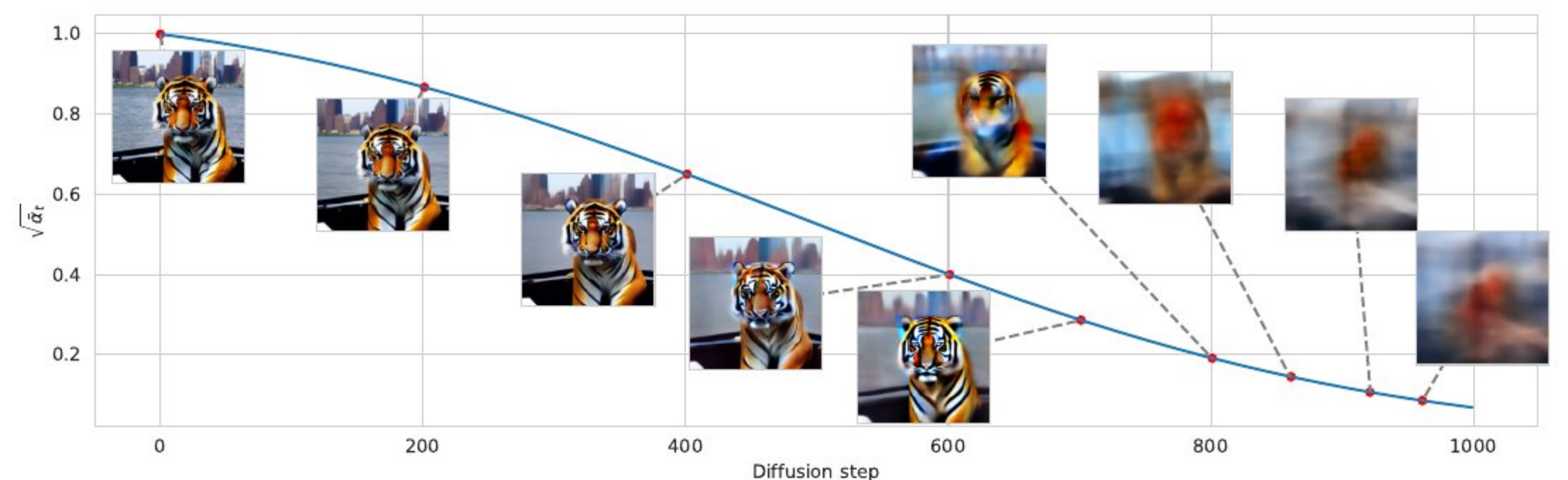


Fig 2. Diffusion of a photo of a tiger on a boat arriving in new york (SD 1.4)



Fig 3. Approach: choose the right step

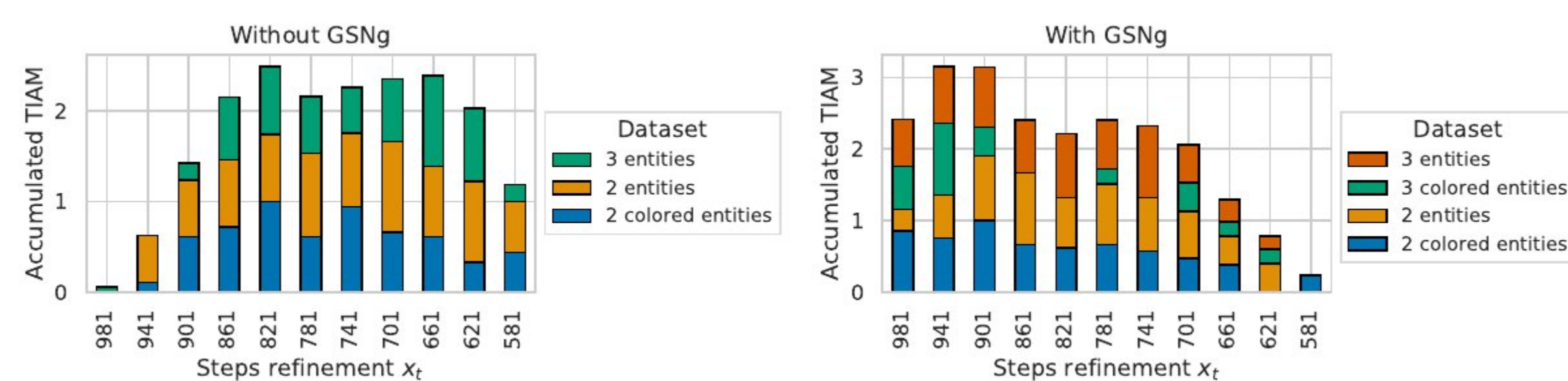


Fig 4. Cumulated TIAM during diffusion without (left) and with GSN (right)

| IterRef | GSNg | Methods          | w/o colors               |                          | with colors              |                         |
|---------|------|------------------|--------------------------|--------------------------|--------------------------|-------------------------|
|         |      |                  | 2 entities               | 3 entities               | 2 entities               | 3 entities              |
| 0       | ✗    | Stable Diffusion | 45.4 <sub>32.2/5.5</sub> | 8.4 <sub>33.5/5.5</sub>  | 3.9 <sub>34.6/5.4</sub>  | 0.1 <sub>34.5/5.4</sub> |
| 1       | ✗    | InitNo           | 62.1 <sub>33.1/5.5</sub> | 14.2 <sub>34.3/5.4</sub> | 7.2 <sub>35.7/5.4</sub>  | 0.2 <sub>35.5/5.3</sub> |
|         |      | Ours             | 65.8 <sub>33.7/5.5</sub> | 23.1 <sub>35.4/5.5</sub> | 8.7 <sub>36.4/5.4</sub>  | 0.4 <sub>36.3/5.4</sub> |
| 3       | ✓    | Divide&Bind      | 69.9 <sub>33.7/5.5</sub> | 33.6 <sub>35.9/5.4</sub> | 11.3 <sub>36.1/5.3</sub> | 0.5 <sub>36.1/5.3</sub> |
|         |      | Attend&Excite    | 71.4 <sub>34.0/5.5</sub> | 32.0 <sub>35.9/5.4</sub> | 10.5 <sub>36.9/5.4</sub> | 0.6 <sub>36.9/5.3</sub> |
|         |      | InitNo+          | 75.0 <sub>34.1/5.5</sub> | 34.2 <sub>36.0/5.4</sub> | 11.9 <sub>37.1/5.4</sub> | 1.0 <sub>37.3/5.3</sub> |
| 0       | ✓    | Syngen           | 78.5 <sub>34.1/5.4</sub> | 39.2 <sub>36.5/5.4</sub> | 20.4 <sub>37.1/5.3</sub> | 2.4 <sub>36.8/5.3</sub> |
| 1       | ✓    | Syngen+          | 75.8 <sub>33.8/5.3</sub> | 36.2 <sub>36.2/5.4</sub> | 20.1 <sub>37.1/5.3</sub> | 1.9 <sub>36.9/5.3</sub> |
|         |      | Ours+            | 81.1 <sub>34.2/5.4</sub> | 45.8 <sub>36.7/5.4</sub> | 20.5 <sub>37.1/5.3</sub> | 2.8 <sub>37.1/5.3</sub> |

Tab 1. Performances (TIAM<sub>CLIP</sub> score/aesthetics score)

InitNO+, Ours+: with GSN



Fig 5. Results without GSN (same seed)

Generalize well across the VQA score [Lin et al. 2024] and GenEval [Ghosh et al. 2023]

## Further aspects and perspectives

- Already applied to Flow Matching models, with comparable results: Stable Diffusion 3
- SAGA: Learning Signal-Aligned Distributions for Improved Text-to-Image Generation (Grimal et al, AAAI 2026)
- Automated selection of refinement steps to target specific aspects of the prompt

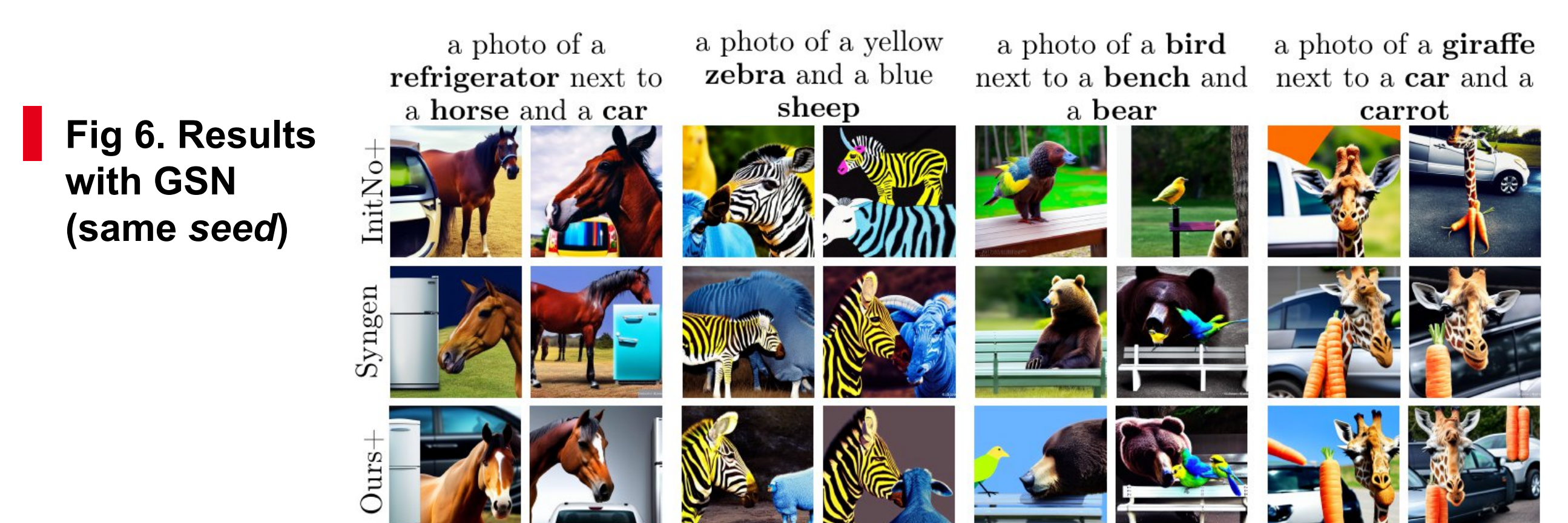


Fig 6. Results with GSN (same seed)